Ahsanullah University of Science and Technology

Department of Electrical & Electronic Engineering

ASSIGNMENT

Course No : **EEE 4105**

Course Name : Control System-I

Date of Submission : 04/10/2026

Submitted by:

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Contribution of Each Member

Assignment on PID Controller Design for a Car Suspension System (Design ID: 20220205057)

SL	Name	Student ID	Contribution
1	Radit Mamnun (Leader)	20220205057	Problem formulation, calculation of A, B and design specifications, damping ratio (ζ) derivation, final compilation and submission
2	Alif Al Noor	20220205050	Uncompensated system analysis: root locus construction, operating point s_1, gain K, peak time and steady-state error
3	Sultan Raonok	20220205028	Desired pole location and PD controller design (angle deficiency, zero location z_c, gain K_d)
4	Kowshik Nahin Tonmoy	20220205189	PI controller design, final PID gain calculation, closed-loop transfer function and pole analysis
5	Rayan Bin Alam	20220205023	MATLAB code development and verification, root locus and step response plots, sisotool design
6	Md. Zahid Hasan	20220105250	Simulink modelling and simulation, results comparison, discussion and conclusion

Q1. Mathematical Design of the PID Controller (Hand Calculations)

ID: 20220205057

Problem Statement:

A cars suspension system is represented by a unity feedback system with a proportional controller K and plant.

$$G(s) = (s+9) / [(s+2)(s+5)(s+11)]$$

We are required to design a PID controller so that the system operates with a peak time that is $(1/4)$ times, but not greater than 70% of, that of uncompensated system operating at 80% overshoot, with zero steady state error for a unit step input.

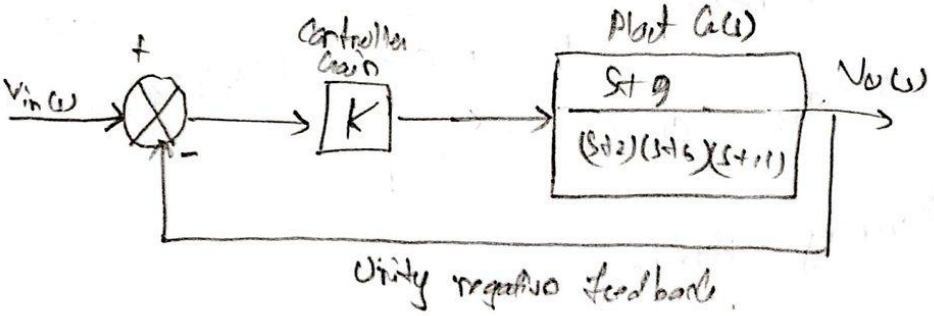


Fig: Uncompensated system

Given,

Open-loop poles : $s = -2, -5, -11$.

Open-loop zero : $s = -9$.

Number of poles $n = 3$, number of zeros $m = 1$.

Determination of A and B

Student ID used: 2022020505 7

Second last digit = 5.

Last digit = 7 (neighbour is zero).

A = next integer of $(5/3)$ = next integer of $2.5 = 3$.

B = next integer of $(7/2)$ = next integer of $3.5 = 4$.

Design specifications:

(i) %OS of uncompensated system = 80% = 40%

(ii) $T_{p,new} = (1/A)\, T_{p,old} = (1/3)\, T_{p,old}$.

(iii) $e_{ss} = 0$ for unit step.

Damping Ratio:

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2[\%OS/100]}}$$

$\Rightarrow \ln(0.40) = -0.9163$;

$\Rightarrow \ln^2(0.40) = 0.8306$; $\pi^2 = 9.8696$.

$\Rightarrow \zeta = 0.9163/\sqrt{(9.8696+0.8306)} = 0.9163/3.2925$

$\therefore \zeta = 0.28$

$\theta = \cos^{-1}(\zeta) = \cos^{-1}(0.28) = 73.74°$

So, the dominant closed-loop poles must lie on a radial line making 73.74° with the negative real axis.

(b) Root Locus Construction Rules:

Number of branches = $n = 3$.

Real-axis segments: $-5 \le \sigma \le -2$ and $-11 \le \sigma \le -9$

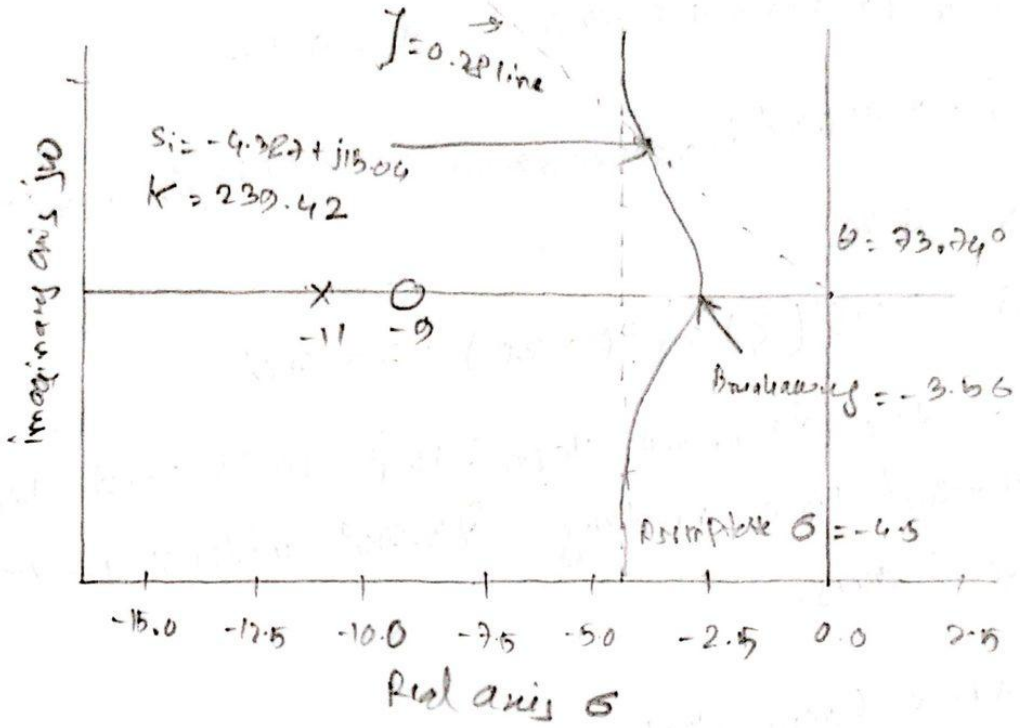
Number of asymptotes = $n - m = 2$.

Asymptote angles: $\theta_a = (2k+1)180°/2 = \pm 90°$.

Centroid: $\sigma_a = [(-2-5-11) - (-9)]/2 = (-18+9)/2$
$= -4.5$.

Two branches start at -2 and -5, break away from the real axis between them and move toward $\pm j\infty$ along the asymptote at $\sigma = -4.5$. The

third branch goes from -11 to the zero at -9.



Locating the Dominant Pole:

Search along the $\zeta = 0.28$ line, $s = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}$.

for the point satisfying the angle condition:

$$\angle(s+9) - \angle(s+2) - \angle(s+5) - \angle(s+11) = \pm 180°.$$

The angle condition is satisfied at $\omega_n = 15.67$ rad/s.

$$s_1 = -4.387 + j15.04.$$

$$\text{Angle} = 72.95° - (99.02° + 89.67° + 66.27°) = -180°.$$

Gain (Magnitude Condition).

$$K = (15.229 \times 15.053 \times 16.430)/15.932$$

$$\Rightarrow K = 239.42.$$

Peak time:

$$\omega_d = 15.04 \text{ rad/s} \therefore T_{p,old} = \pi/\omega_d = 3.1416/15.04$$

$$\Rightarrow T_{p,old} = 0.2089 \text{ s}$$

Steady state error:

$$K_p = \lim_{s \to 0} KG(s) = 239.42 \times 9/(205 \times 11) = 10.59.$$

$$e_{ss} = 1/(1+K_p) = 1/20.59 = 0.0486 \ (4.86\%).$$

Q) Desired (Compensated) Dominant Pole:

Required Peak Time:

$$T_{p,new} = (1/3) \times T_{p,old} = (1/3) \times 0.2089.$$

$$T_{p,new} = 0.0696 \text{ s}.$$

Location of Desired Pole:

$$\omega_{d,new} = \pi/T_{p,new} = 3.1416/0.0696 = 45.12 \text{ rad/s}.$$

damping ratio:

The overshoot must remain the same. so [20:28

$\sigma_{new} = \omega_{d,new} / \tan(73.74°) = 45.12/3.428 = 13.16.$

$\omega_{n,new} = \sqrt{(13.16^2 + 45.12^2)} = 47.0$ rad/s.

$S_{new} = -13.16 + j45.12.$

The point S_{new} does not lie on the compensated root locus. Hence the required response cannot be achieved by adjusting gain K alone. A PD (ideal derivative) compensator is used to add a zero that reshapes the root locus to pass through S_{new}.

PD compensator: $G_{PD}(s) = K_d(s+z_0)$.

Angle of 2nd quadrant vector $= 180° - \tan^{-1}\left(\frac{Im}{Re}\right)$

eg: Zero - D: $180° - \tan^{-1}\left(\frac{46.122}{4.161}\right) = 95.27°.$

$\angle$ angle $= 95.27° - (103.80° + 100.25° + 92.74°)$

$= -201.62°.$

ID: 20220205057

Required compensation angle:

$\theta_c = -180° - (-201.62°)$

$\theta_c = 21.62°.$

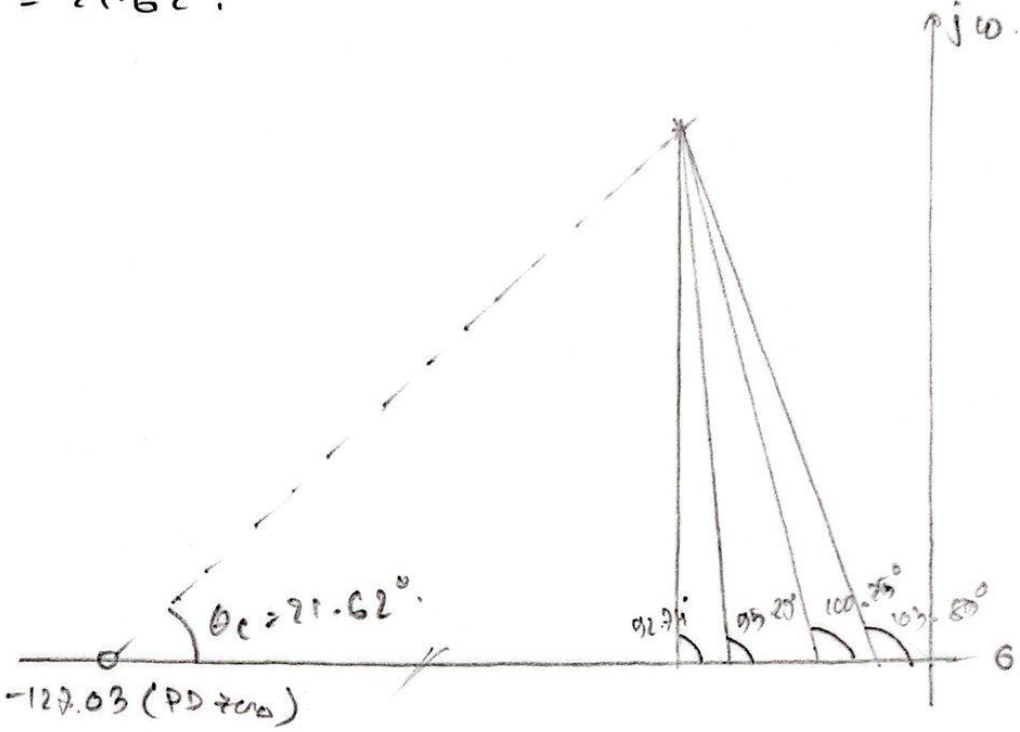


fig: Angle contribution at s-need.

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PD Zero Location and Gain

- **Location of PD zero**

$$\tan\theta_c = \frac{\omega_{d\,new}}{(z_c - \sigma_{new})}$$

$$\Rightarrow \tan(21.62^\circ) = \frac{45.12}{(z_c - 13.16)}$$

$$\Rightarrow z_c - 13.16 = \frac{45.12}{0.3963} = 113.87$$

$$\therefore z_c = 127.03$$

$\therefore$ PD zero at $s = -127.03$.

- **Gain K_d**

$$K_d = \frac{(|s+2||s+5||s+11|)}{(|s+9||s+127.03|)} \quad \text{at } s = s_{new}$$

$$|s_{new} + 127.03| = |113.865 + j45.122|$$

$$= 122.48$$

$$\therefore K_d = \frac{(46.482 \times 45.854 \times 45.174)}{(45.314 \times 122.48)}$$

$$\Rightarrow K_d = \frac{96283.21}{5550.1}$$

$$\therefore K_d = 17.35$$

- **PD Controller**

$$G_{PD}(s) = 17.35\,(s + 127.03)$$

$$= 17.35s + 2203.71$$

- **Root locus after PD compensation**

Poles: $-2, -5, -11$

zeros: $-9, -127.03$

$n - m = 1$, so one asymptote at $180°$.

Real-axis segments: $[-5, -2]$, $[-11, -9]$ and $\sigma \leq -127.03$. The complex branches now bend to the left and pass through s_{new}, then come back to the real axis to the left of -127.03.

ess with PD only: $K_p = 17.35 \times 127.03 \times \frac{9}{110}$

$$= 180.3$$

$$\therefore \quad e_{ss} = \frac{1}{180.3} = 0.0055 \neq 0$$

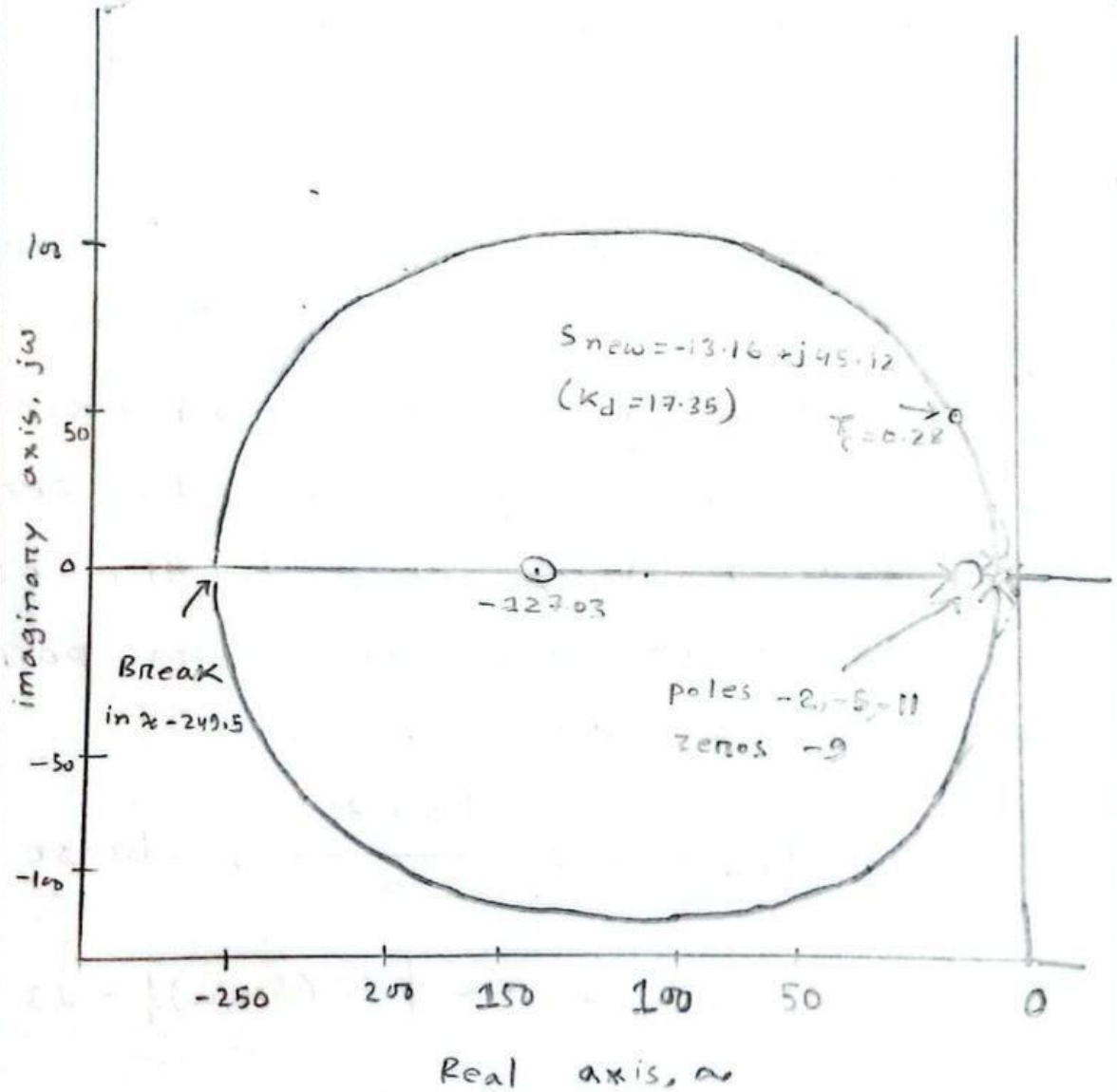


Fig: Root locus after PD compensation

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PI Controller Design

With PD compensation, the system is still Type 0 (no pole at origin), so a finite steady-state error (0.0055) remains. To make $e_{ss} = 0$ for a step input, an integrator (pole at $s = 0$) must be added in the forward path. To avoid disturbing the transient response already designed, the PI zero is placed very close to the origin. so that the pole-zero pair nearly cancels at s_{new}.

$$G_{PI}(s) = \frac{(s + z_i)}{s}, \text{ choose } z_i = 0.1$$

$$z_i = 0.1 \ll |Re(s_{new})| = 13.16$$

So, angle added at s_{new},

$$\angle(s_{new} + 0.1) - \angle(s_{new}) \approx 106.15^\circ - 106.26^\circ$$
$$= -0.11^\circ \text{ (negligible)}$$

$$\therefore G_{PI}(s) = \frac{(s+0.1)}{s}$$

- **Steady-state error with PI**

$$K_p = \lim_{s \to 0} G_c(s)\, G(s)$$

$$= \lim_{s \to 0} \frac{\left[17.35\,(s+127.03)\,(s+0.1)\,(s+9)\right]}{\left[s\,(s+2)\,(s+5)\,(s+11)\right]}$$

$$= \infty$$

$$e_{ss} = \frac{1}{(1+\infty)} = 0$$

Final PID controller and Closed-Loop TF

- **PID controller**

$$G_c(s) = G_{PD}(s) \cdot G_{PI}(s)$$

$$= \frac{K_d\,(s+z_c)\,(s+z_i)}{s}$$

$$= K_d s + K_d(z_c + z_i) + \frac{K_d z_c z_i}{s}$$

comparing with $G_c(s) = K_p + \frac{K_i}{s} + K_d s$,

$K_d = 17.35$

$K_p = 17.35 \times (127.03 + 0.1)$

$= 17.35 \times 127.13 = 2205.45$

$K_i = 17.35 \times 127.03 \times 0.1 = 220.37$

$\therefore G_c(s) = 2205.45 + \frac{220.37}{s} + 17.35s$

- Closed - loop transfer function

Numerator:

$17.35 (s + 127.03)(s + 0.1)(s + 9)$

$= 17.35 (s^3 + 136.13 s^2 + 1256.87s + 114.33)$

$= 17.35 s^3 + 2361.9 s^2 + 20072s + 1983.6$

Denominator of $G_c G$:

$s(s+2)(s+5)(s+11) = s^4 + 18s^3 + 87s^2 + 110s$

$$T(s) = \frac{(17.35s^3 + 2361.9s^2 + 20072s + 1983.6)}{(s^4 + 35.35s^3 + 2448.9s^2 + 20182s + 1983.6)}$$

Closed-loop poles: $-13.11 \pm j45.13$, -9.03, -0.099

The dominant poles are almost exactly at the design point. The pole at -9.03 nearly cancels the zero at -9 and the pole at -0.099 nearly cancels the zero at -0.1.

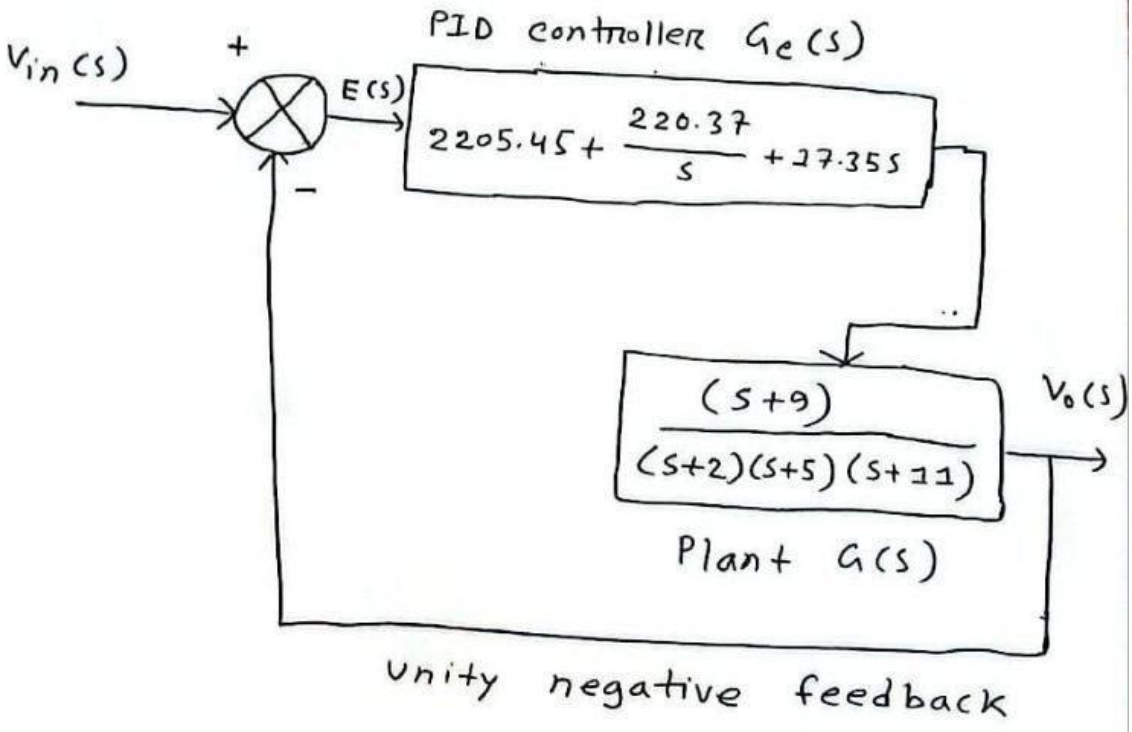


Fig: PID-compensated system.

Results and comparison:

comparison of performance :

Parameter	Uncompensated	PD	PID	Requirement
controller	K=239.92	17.35(s+127.03)	2205.95+220.37/s +17.35s	-
Dominant poles	-4.99±j15.09	-13.16± j45.12	-13.11 ± j45.13	-13.16 ± j45.12
Peak Time T_p	0.2076 s	0.0612 s	0.0612 s	≤0.0696 s
%OS	41.4 %	43.4%	42.8 %	≈ 40 %
Settling time (2%)	0.883 s	0.287 s	0.294 s	-
ess (step)	0.1486	0.0055	0	0

peak time ratio: 0.0612/0.2076 = 0.295 ≈ 1/3.4
(<1/3 and <0.20)

Discussion:

1] The PD controller moved the dominant poles from -4.99± j 15.09 to -13.16± j45.12, which is three times further from the origin on the same ∠ line. This reduced the peak time by about 3.4 times and the settling time by about 5 times.

2] The PI part added an integrator, making the system Type 1, so the steady-state error became exactly zero. Since it's zero (-0.1) is very close to it's pole (0), the transient response was almost unchanged.

3] The simulated overshoot (42.8%) is slightly higher than the design value of 40%, and the peak time (0.0612 s) is slightly smaller than the predicted 0.0696 s. This is because the design assumed a pure second-order system, whereas the actual closed-loop system has extra zeros at −127.03, −9 and −0.1 and extra poles at −9.03 and −0.099. The zeros make the response faster and increase the overshoot slightly. The difference is small, so the second-order approximation is acceptable.

4] The peak time requirement is met with margin, and the steady-state error is met exactly.

5] The hand calculations, MATLAB, sisotool and Simulink results all agree, which confirms the correctness of the design.

6] In a practical suspension system, the pure derivative term amplifies high-frequency noise from road disturbances and sensors. Therefore a filtered derivative ($N = 100000$ in Simulink) should be used in implementation.

ID: 20220205189

conclusion: A PID controller was designed for the car suspension system using the root locus method and verified with MATLAB, sisotool, and simulink. Using ID 20220205057 (A=3, B=4), the uncompensated system at 40% overshoot had a peak time of 0.2089 s and a steady state error of 0.0486. The designed controller

$$G_c(s) = 2205.45 + 220.37/s + 19.35s$$

reduced the peak time to 0.0612 s (less than 1/3 of the original and well within the 70% limit), kept the overshoot close to 40%, and eliminated the steady-state error for a 8 unit step input. Hence all design specifications are satisfied.

Q2. PID Controller Design Using Modern System Tools

2.1 MATLAB Code (PID_Design.m)

```
%% EEE4105 Control System-I - PID Controller Design (Car Suspension)
% Design ID: 20220205057  ->  A = 3, B = 4
clc; clear; close all;
s = tf('s');
G = (s+9)/((s+2)*(s+5)*(s+11))            % Plant

%% Step 1: A, B and specifications
A = ceil(5/2)                              % second-last digit = 5
B = ceil(7/2)                              % last digit = 7
OS = B*10                                  % percent overshoot = 40
factor = min(1/A, 0.7)                     % peak-time factor = 1/3

%% Step 2: damping ratio
zeta = -log(OS/100)/sqrt(pi^2 + log(OS/100)^2)
theta_deg = acosd(zeta)

%% Step 3: uncompensated system (root locus point on zeta line)
sx = @(wn) -zeta*wn + 1j*wn*sqrt(1-zeta^2);
g  = @(wn) mod(angle(evalfr(G, sx(wn))), 2*pi) - pi;   % angle condition
wn1 = fzero(g, [10 20])
s1  = sx(wn1)
K   = 1/abs(evalfr(G, s1))                 % magnitude condition
Tp_old = pi/imag(s1)
T_unc = feedback(K*G, 1);
disp('---- Uncompensated system ----'); stepinfo(T_unc)
ess_unc = 1 - dcgain(T_unc)

figure(1); rlocus(G); sgrid(zeta, 0); grid on;
title('Uncompensated Root Locus with \zeta = 0.28 line');

%% Step 4: desired compensated pole
s_new  = s1/factor
Tp_new = pi/imag(s_new)

%% Step 5: PD compensator
phi = mod(pi - angle(evalfr(G, s_new)), 2*pi);  % angle the PD zero must add
phi_deg = rad2deg(phi)
zc = imag(s_new)/tan(phi) - real(s_new)
Kd = 1/abs(evalfr((s+zc)*G, s_new))
T_pd = feedback(Kd*(s+zc)*G, 1);
disp('---- PD compensated system ----'); stepinfo(T_pd)

figure(2); rlocus((s+zc)*G); sgrid(zeta, 0); grid on;
title('PD-Compensated Root Locus');

%% Step 6-7: PI compensator and final PID
zi = 0.1;
Kp = Kd*(zc + zi)
Ki = Kd*zc*zi
C  = pid(Kp, Ki, Kd)
T  = feedback(C*G, 1)
disp('---- PID compensated system ----'); stepinfo(T)
ess_pid = 1 - dcgain(T)
closed_loop_poles = pole(T)

%% Step 8: step response comparison
figure(3); step(T_unc, T_pd, T, 1); grid on;
legend('Uncompensated', 'PD', 'PID');
title('Unit Step Response Comparison');
```

2.2 MATLAB Results

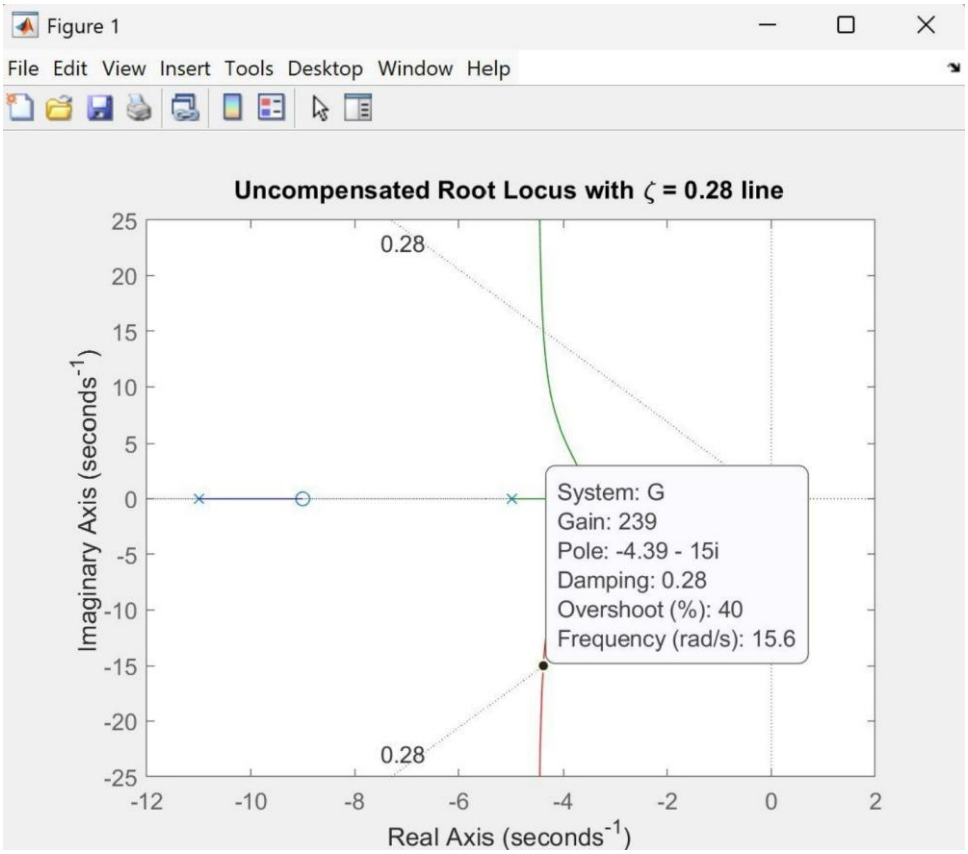


Figure 1: MATLAB – Uncompensated root locus with ζ = 0.28 line (K = 239, 40% overshoot, $s_1 \approx -4.39 \pm j15$)

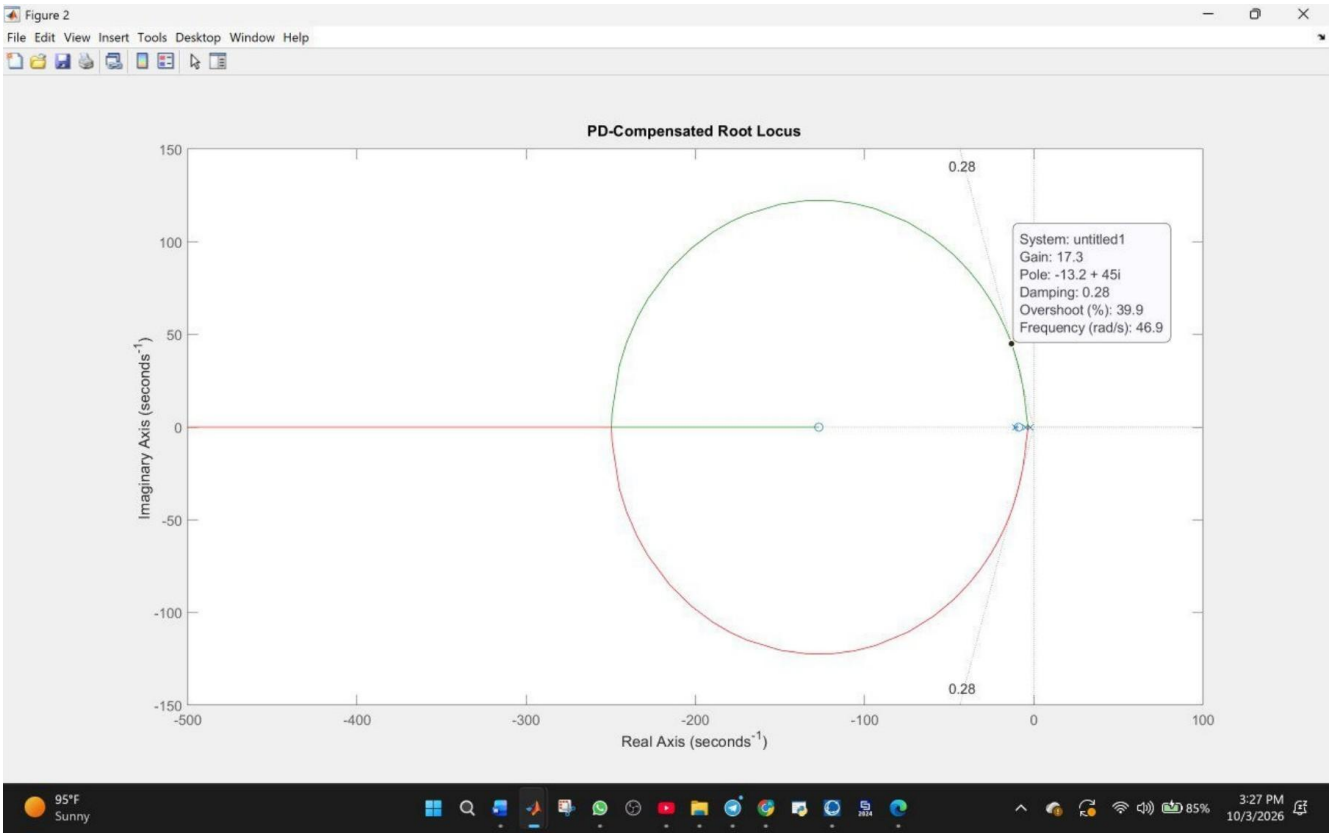


Figure 2: MATLAB – PD-compensated root locus passing through s_new ≈ −13.2 + j45 (gain = 17.3 = K_d)

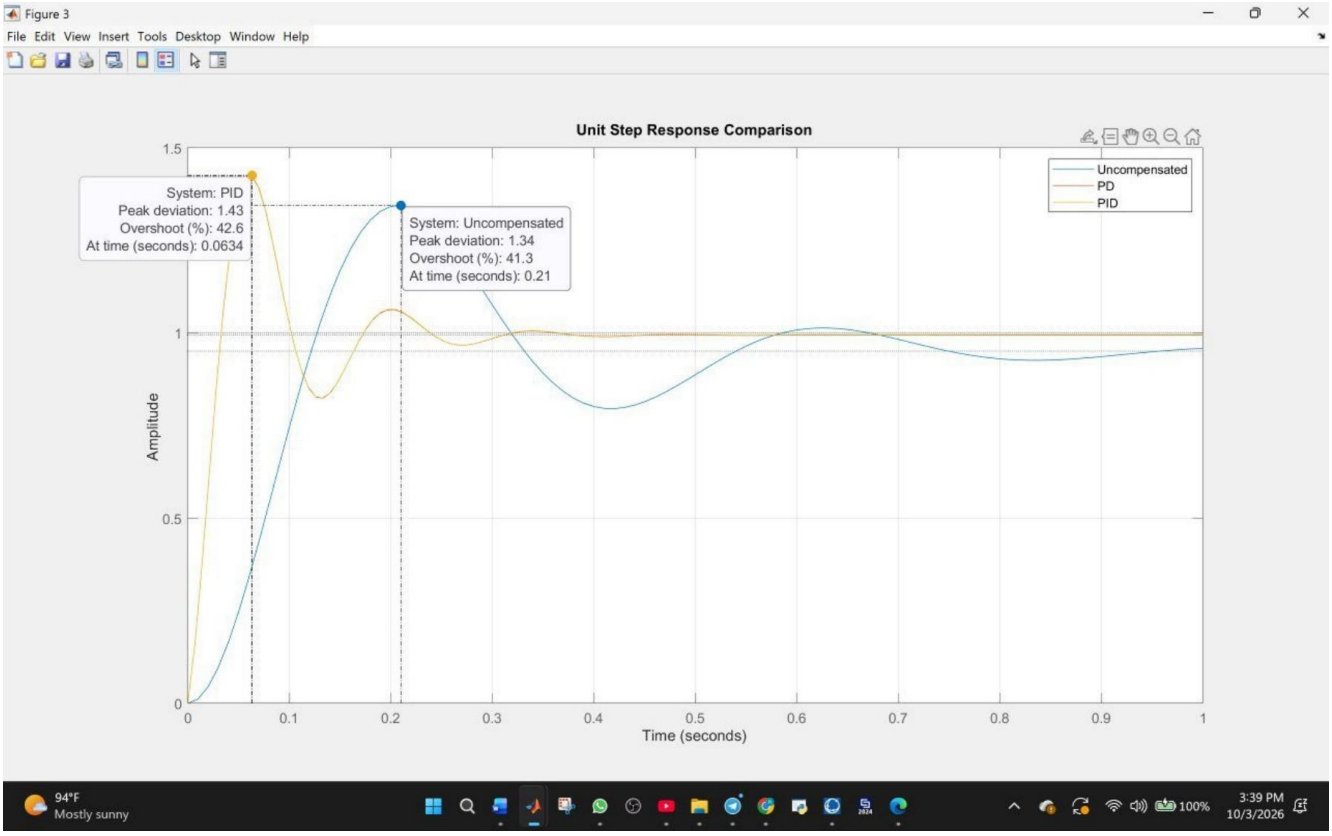


Figure 3: MATLAB – Unit step response comparison: uncompensated (Tp = 0.21 s, OS = 41.3%) vs. PID (Tp = 0.063 s, OS = 42.6%)

2.3 Design Using sisotool (Control System Designer)

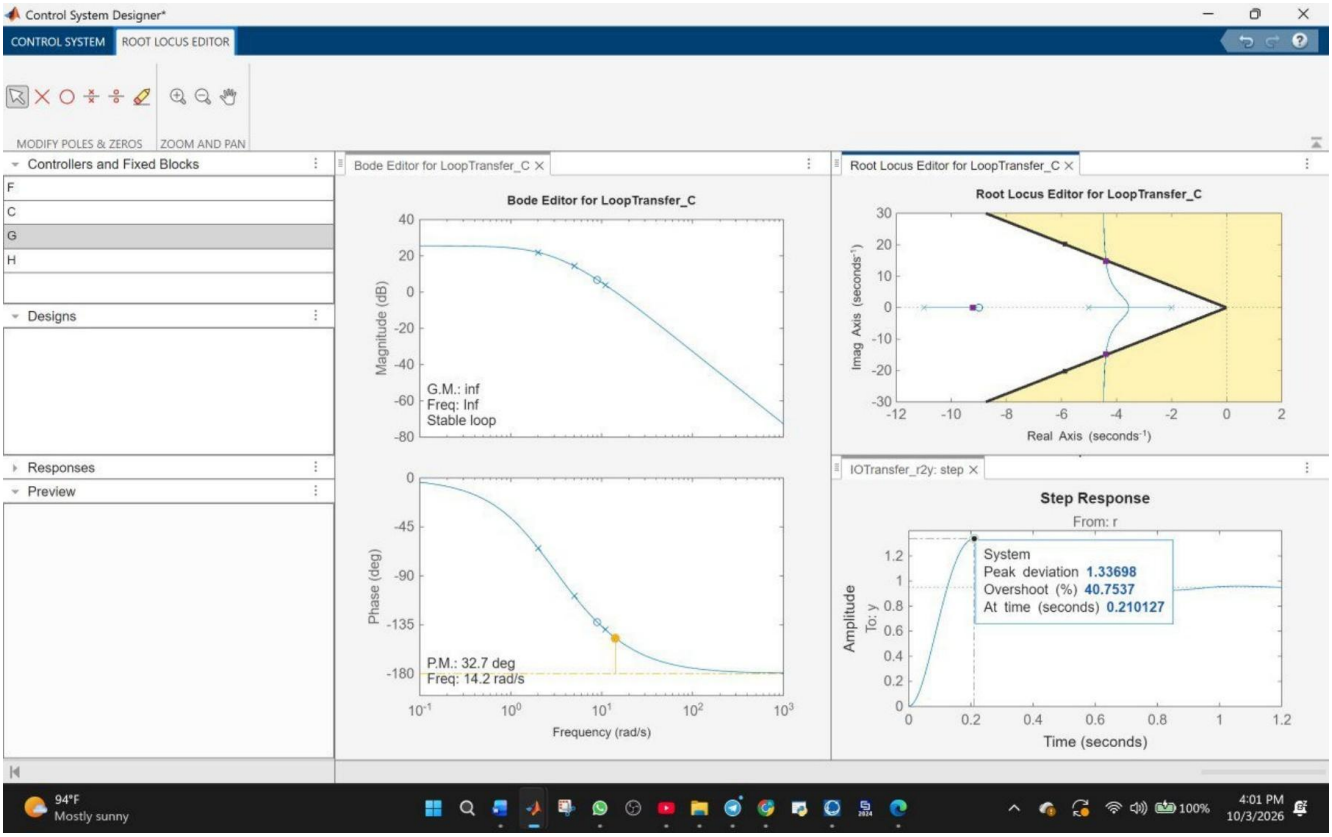


Figure 4: sisotool – Uncompensated design: Bode editor, root locus editor with 40% overshoot requirement, and step response

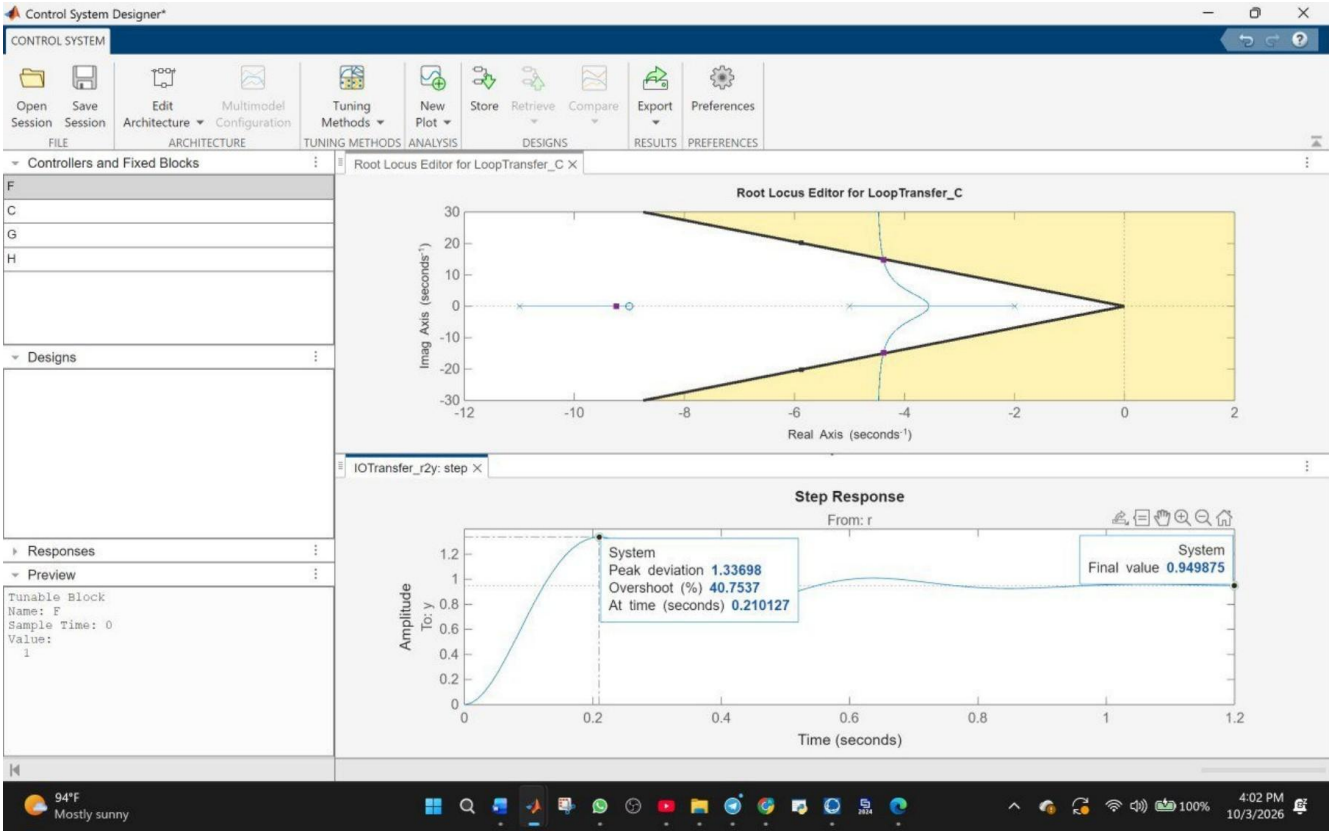


Figure 5: sisotool – Uncompensated root locus and step response (Tp = 0.210 s, OS = 40.75%, final value = 0.950)

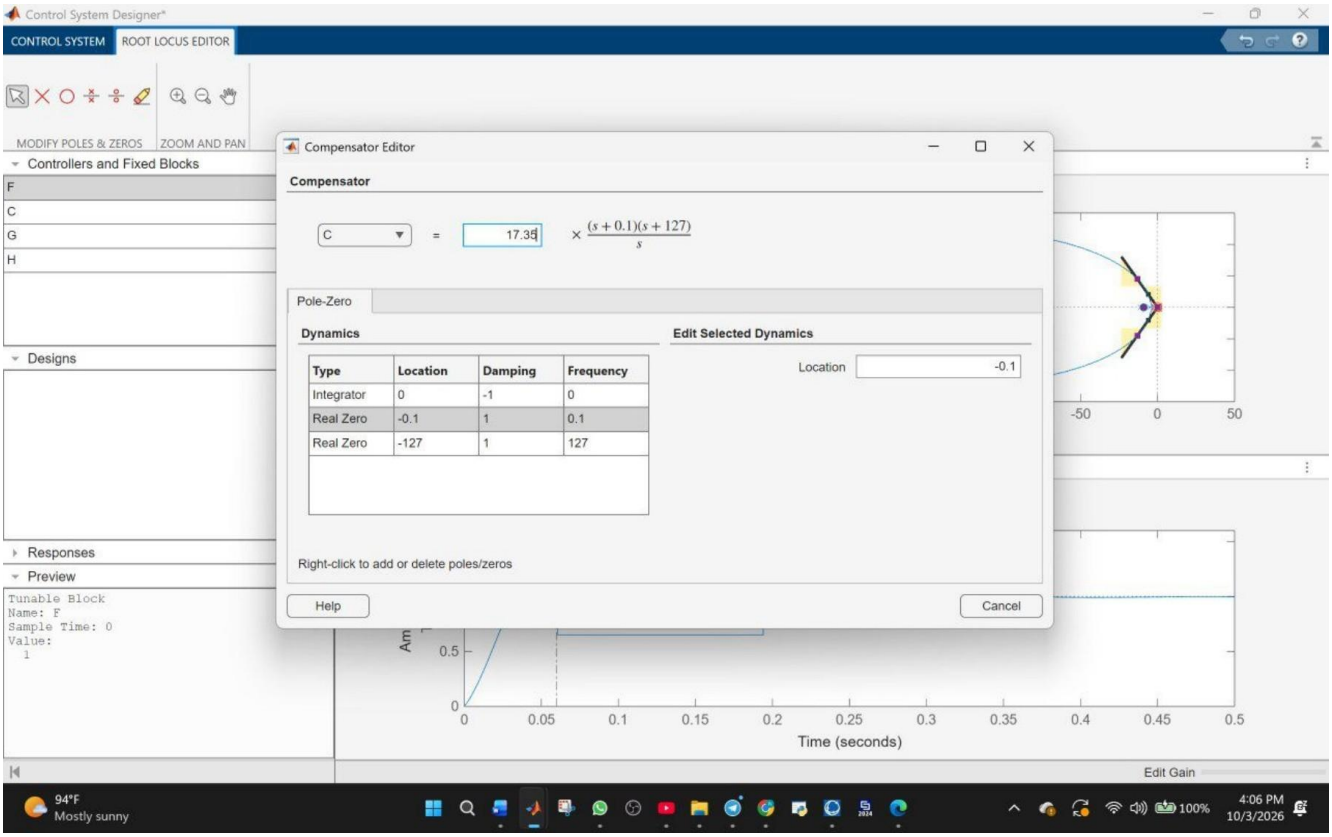


Figure 6: sisotool – Compensator Editor: C(s) = 17.35 (s + 0.1)(s + 127) / s (integrator + two real zeros)

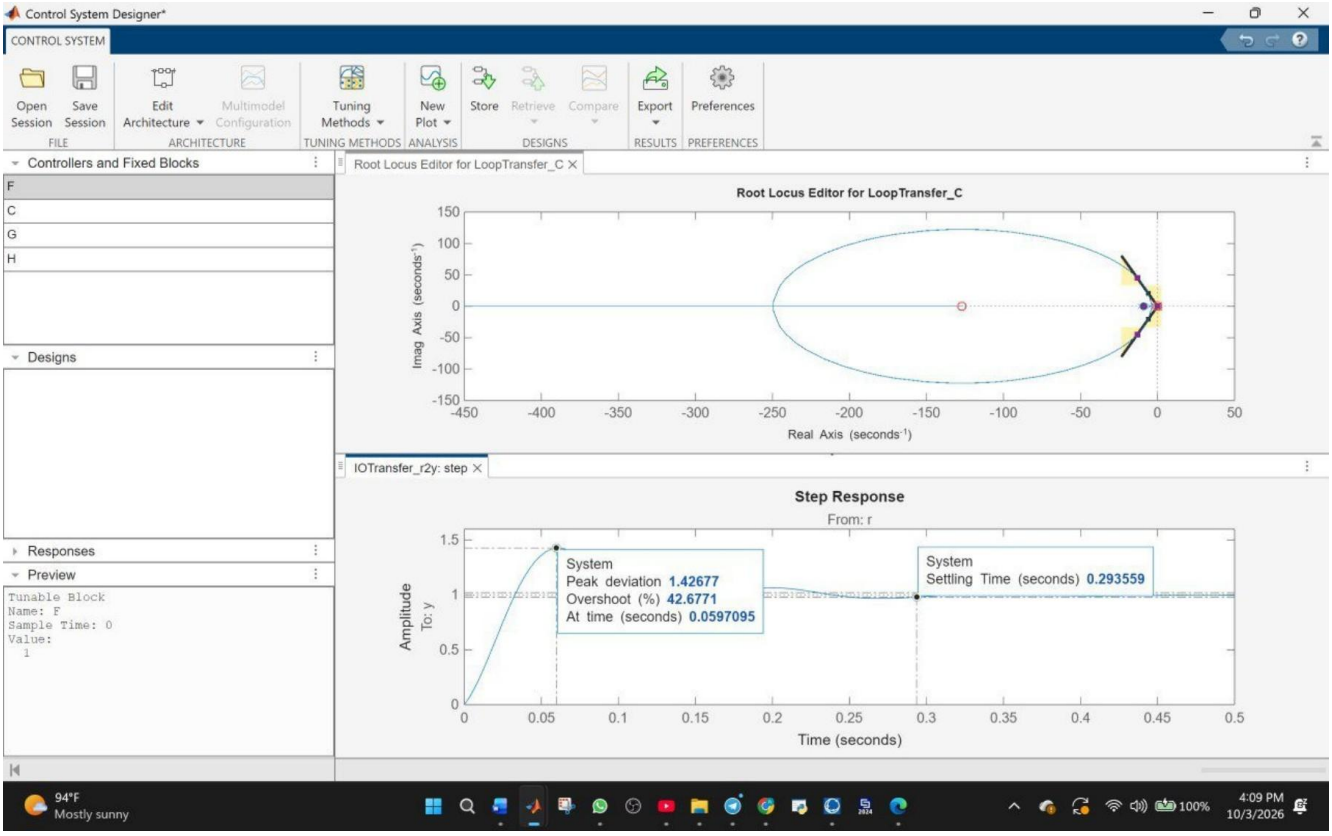


Figure 7: sisotool – PID-compensated root locus and step response (Tp = 0.0597 s, OS = 42.68%, Ts = 0.294 s, zero steady-state error)

2.4 Simulation Using Simulink

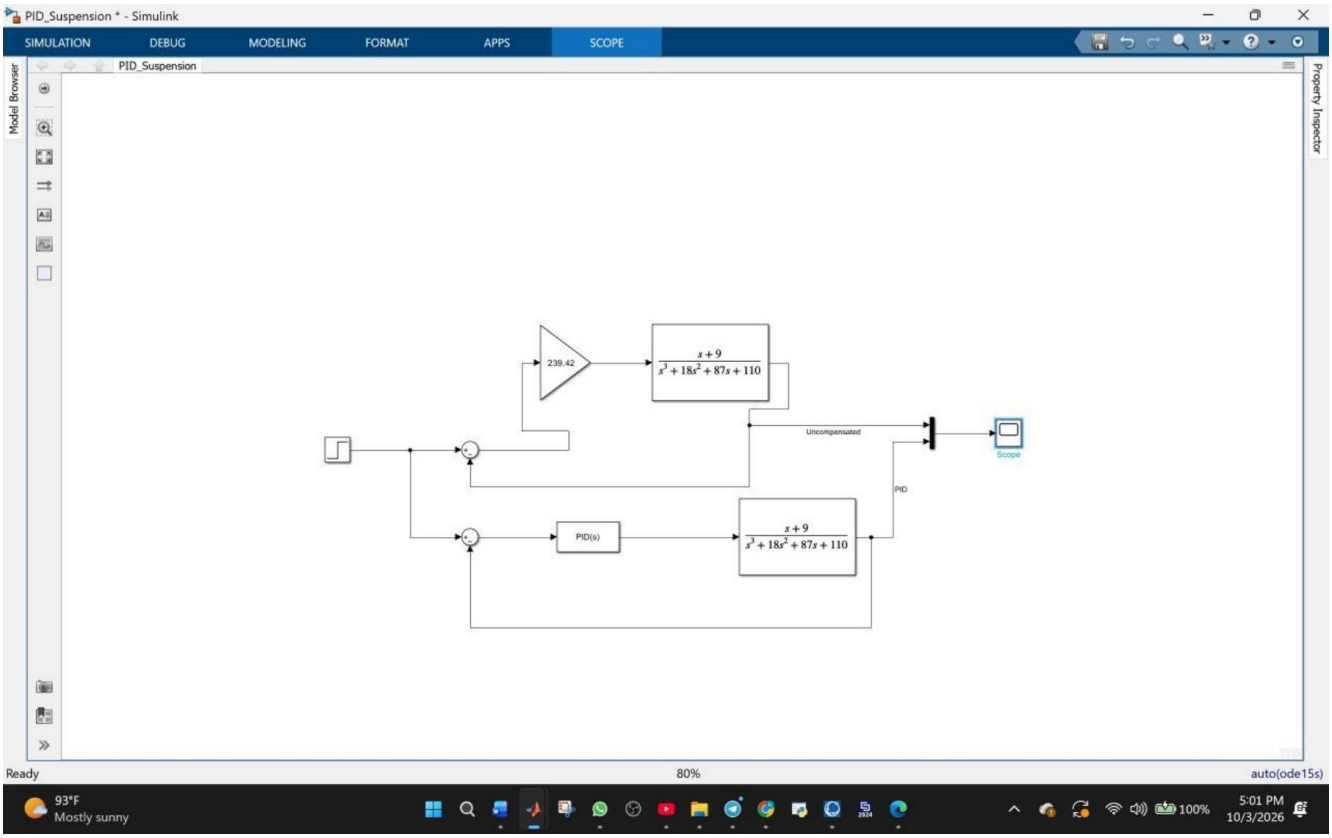


Figure 8: Simulink – Model PID_Suspension.slx with uncompensated loop (K = 239.42) and PID-compensated loop

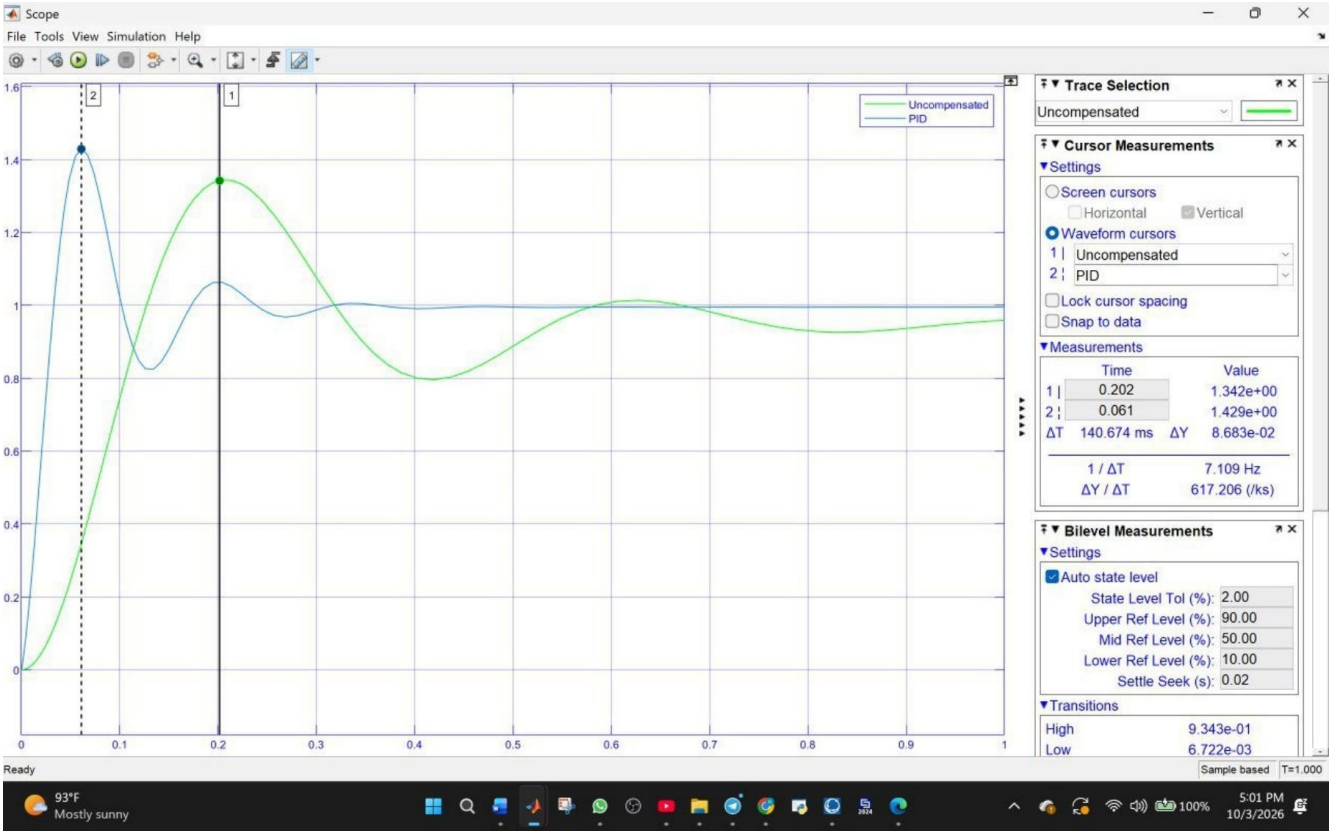


Figure 9: Simulink – Scope output with cursor measurements: PID peak 1.429 at 0.061 s, uncompensated peak 1.342 at 0.202 s